Model Card

Bayesian-network outreach ranking and Bayesian intersectional audit
ACSPublicCoverage, California 2018 (ICT 5143 Capstone, Problem 8)

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Version 1.0 · results from the executed notebook of 25 October 2026

Read first. $Y = 1$ means *no public health coverage* in the ACS person record. It does **not** mean uninsured and it does **not** mean eligible for any programme. $D = 1$ means selection for *hypothetical* information outreach. This model is a research artefact for studying fairness audits. It must not be used to make or support decisions about real people. All numbers come from `Problem8_Intersectional_Fairness.ipynb` with fixed seeds. Synthetic and semi-synthetic results describe the procedures, not the California population.

1 Model details

- **Developers:** Sumedh Navuda and Aldino, ICT 5143 Algorithms, Fairness and Graphical Models Laboratory. Contact: `sumedh.mitmpl2026@learner.manipal.edu`, `aldino.mitmpl2026@learner.manipal.edu`. Repository: `https://github.com/sumedhnvda/ict5143-p8-fairness-audits`.
- **Date and version:** v1.0, 25 October 2026. Code: `Problem8_Intersectional_Fairness.ipynb` (self-contained, executed); environment pinned in `requirements-lock.txt`.
- **Predictive model:** a discrete Bayesian network with an augmented naive-Bayes structure (Figure 1). Y is a parent of all 18 features, plus the feature edges SCHL|AGEP, ESR|AGEP, PINCP|ESR, DIS|AGEP, MAR|AGEP, CIT|NATIVITY, {DEAR, DEYE, DREM}|DIS and FER|SEX. ST is constant within one state and is omitted. The CPTs are Dirichlet-smoothed ($\alpha = 1$), which makes every entry positive. AGEP and SCHL use fixed bins, and PINCP uses training-split quartiles. **The graph is associational, not causal.**
- **Inference:** exact. $P(Y \mid \text{evidence})$ by variable elimination. Our VE under two elimination orders, pgmpy VE, pgmpy clique-tree belief propagation and enumeration agree to 3.3×10^{-16} over 200 random evidence masks. The vectorised predictor equals general VE, and the joint CI SEX $\perp$ {AGEP, DIS} | Y holds in the fitted model (notebook §3.1).
- **Decision rule:** $D = 1$ if the predicted probability is in the top 30% of validation scores (group-unaware, $\lambda = 0$). Fairness-adjusted variants $R(\lambda)$ blend the global threshold with group-specific equal-opportunity thresholds: $t_g(\lambda) = (1-\lambda)t_{\text{global}} + \lambda t_g^{\text{EOpp}} + c$, with c set so that exactly 30% of validation records are selected. λ is chosen on validation only (notebook §10).
- **Baselines:** logistic regression (one-hot categories plus scaled AGEP, SCHL, PINCP) and histogram gradient-boosted trees (300 iterations, learning rate 0.08), with the same information and splits.
- **Audit models** (for $k_g \sim \text{Bin}(n_g, \theta_g)$, where n_g = positives and k_g = selected positives in intersection g): raw k/n; independent Beta(1,1); complete pooling; empirical-Bayes beta-binomial (hier); full-Bayes beta-binomial on a grid; the attribute-structured hierarchical logistic audit (struct), with logit $\theta_g = \beta_0 + \beta_{\text{race}} + \beta_{\text{sex}} + \beta_{\text{dis}} + u_g$ and $u_g \sim N(0, \tau^2)$; and the **proposed** ϵ-contamination prior $\theta_g \sim \pi\,\text{Beta}(a,b) + (1-\pi)\,\text{Beta}(1,1)$ with $\pi \le \pi_{\max}$ (Figure 2). The struct model is a documented restricted version of a hierarchical intersectional audit in the spirit of Foulds et al. It is *not* an exact reproduction of their model.
- **Calibrated audit procedure (frozen in notebook §9):** contamination prior with $\pi_{\max} = 0.8$ and the plug-in rule "declare ΔEOpp $\le \epsilon$ if the max−min of the posterior mean TPRs is $\le \epsilon$". It was chosen by a preregistered Youden criterion on synthetic data before any real-data selection.

- **Recommended procedure (notebook §15):** *design-matched calibration* of the contamination audit (slab type, $\pi_{\max}$ and rule chosen by the same Youden criterion on semi-synthetic worlds built from the deployment population's validation group sizes), plus *held-out certification* (choose λ on half of the validation households, certify on the other half). Chosen configurations: CA, fixed slab with $\pi_{\max} = 0.99$ and the upper-0.9 rule; TX, centred slab with $\pi_{\max} = 0.95$ and the upper-0.5 rule.
- **Licence:** MIT licence for our code (`LICENSE`). The data are public-domain ACS PUMS and are not redistributed.
- **Citation:** S. Navuda and Aldino. *When does pooling make intersectional fairness audits falsely reassuring?* ICT 5143 capstone report, Problem 8, 2026. `https://github.com/sumedhnvda/ict5143-p8-fairness-audits`.

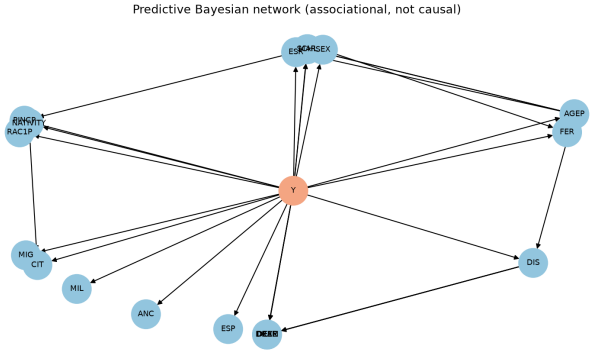


Figure 1: Predictive BN structure (associational).

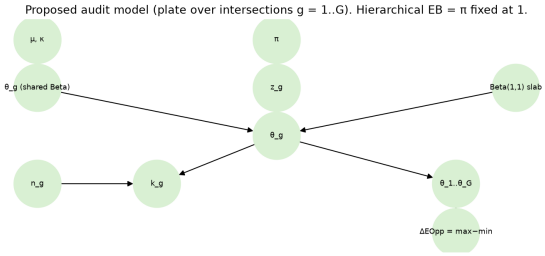


Figure 2: Proposed audit model (plate over intersections). Hierarchical EB is the case $\pi = 1$.

2 Intended use

- **Primary use:** a benchmark for studying when Bayesian intersectional fairness audits are reliable, in particular whether partial pooling hides a small disadvantaged intersection and whether validation-time tolerance claims hold on independent data.
- **Primary users:** fairness auditors, researchers and course staff. The stakeholders named in the problem are low-income residents, public-health outreach teams and community organisations.
- **Out of scope:** deciding who is contacted, enrolled or denied anything; inferring a person's insurance status or programme eligibility; any causal statement about race, sex or disability; use for another state, year or population without re-validation and re-calibration (§8); certifying that a rule is "fair".

3 Factors

- **Protected attributes as recorded:** RAC1P (nine source codes, not collapsed), SEX (1 male, 2 female) and DIS (1 with a disability, 2 without). These are self- or proxy-reported survey categories and limited measurements of identity. Nothing is inferred from names or other variables.
- **Intersections:** race × sex × disability gives 36 cells, of which 35 are non-empty in the seed-0 test split. Two-attribute groupings (race×sex, sex×disability, race×disability) and single attributes are reported as baselines.
- **Small groups:** cells with fewer than 30 positives or negatives are flagged exploratory and reported, never dropped. In the seed-0 test split, 14 of the 35 non-empty intersections have fewer than 30 positives, and one (`7|2|1`) has none.
- **Use of protected attributes at prediction time:** all three models use RAC1P, SEX and DIS as features, as in the Folktables definition. Rules with $\lambda > 0$ also use group-specific thresholds.

4 Metrics

- **Primary predictive metric:** recall at 30% outreach capacity. Also AUROC, average precision, precision at capacity, log loss, Brier score, per-group expected calibration error and worst-group recall.
- **Primary fairness metric:** intersectional equality of opportunity, $\Delta\text{EOpp} = \max_g t_g - \min_g t_g$ over intersections with ≥ 30 positives, with tolerance $\epsilon = 0.10$ (robustness: 0.05, 0.15). The diagnostics are ΔDP, ΔEOdds, the selection-rate ratio R_{sel} and the worst-group TPR.
- **Model-based audit metrics:** posterior intervals for ΔEOpp, $P(\Delta\text{EOpp} > \epsilon \mid \text{data})$, and in simulation the false tolerance-satisfaction rate (FTS = P(declared fair | truly unfair)), the correct acceptance rate (ACC = P(declared fair | truly fair)), interval coverage and width.
- **Uncertainty:** *evaluation* uncertainty from 1,000 household-cluster (Poisson) bootstrap replicates of the seed-0 test split; *training* uncertainty from 5 refits on independent household splits. They are reported separately.
- **Acceptable utility loss (preregistered):** at most 0.02 recall relative to the unconstrained rule.

5 Training and evaluation data

- **Source:** US Census Bureau, American Community Survey PUMS 2018 1-Year person file for California, obtained through `folktables.ACSDataSource(...).get_data(download=True)`.
- **Task definition:** Folktables `ACSPublicCoverage` (`public_coverage_filter`: AGEP < 65 and PINCP $\leq 30{,}000$), giving 138,554 people in 85,182 households. Our features are identical, cell by cell, to Folktables' `df_to_numpy` output. Label $Y = 1 - \mathbb{1}[\text{PUBCOV} = 1]$ (exact reversal), with prevalence 0.631.
- **Splits:** household-disjoint 60/20/20 (train/validation/test) on SERIALNO, five split seeds. Bins, CPTs, thresholds, λ and audit hyperparameters use training and validation data only. The test split is used only to audit frozen rules.
- **Bookkeeping columns, not features:** SERIALNO (splits and clustered bootstrap); PWGTP and PWGTP1–80 (survey-weighted estimates, notebook §13).
- **Replication population:** Texas 2018 (98,928 people), with identical definitions (notebook §14).

6 Quantitative analyses

6.1 Predictive performance and unconstrained-rule fairness

Table 1: Test metrics of the unconstrained top-30% rule, mean ± sd over 5 household splits.

Metric	BN	LR	GBT
AUROC	0.729 ± 0.004	0.716 ± 0.004	0.768 ± 0.003
AP	0.803 ± 0.002	0.787 ± 0.002	0.836 ± 0.002
Recall@30%	0.388 ± 0.005	0.384 ± 0.006	0.409 ± 0.006
Prec.@30%	0.824 ± 0.004	0.809 ± 0.004	0.860 ± 0.002
Log loss	0.577 ± 0.004	0.584 ± 0.002	0.541 ± 0.003
Brier	0.196 ± 0.002	0.199 ± 0.001	0.182 ± 0.001
max group ECE	0.110 ± 0.010	0.069 ± 0.011	0.063 ± 0.013
ΔEOpp	0.719 ± 0.018	0.625 ± 0.049	0.585 ± 0.019
ΔDP	0.655 ± 0.016	0.548 ± 0.021	0.508 ± 0.013
worst recall	0.000 ± 0.000	0.000 ± 0.000	0.008 ± 0.011

The BN is roughly at the level of logistic regression (AUROC 0.729 vs 0.716). Gradient-boosted trees are best (0.768). Predictive accuracy is not the contribution. The BN's worst per-group calibration error (0.110) is larger than LR's (0.069) and GBT's (0.063); see Figure 3.

6.2 Single-attribute vs intersectional audit

Table 2: Seed-0 test split, unconstrained rule. Gaps use groups with ≥30 relevant observations (adequate/all).

		BN			LR			GBT		
Grouping	adequate/all	ΔEOpp	ΔDP	worst rec.	ΔEOpp	ΔDP	worst rec.	ΔEOpp	ΔDP	worst rec.
SEX	2/2	0.162	0.128	0.291	0.080	0.060	0.338	0.095	0.077	0.346
DIS	2/2	0.384	0.324	0.021	0.396	0.334	0.010	0.354	0.307	0.067
RACE	8/9	0.597	0.548	0.079	0.561	0.456	0.106	0.360	0.313	0.169
SEX×DIS	4/4	0.468	0.386	0.009	0.436	0.361	0.004	0.427	0.352	0.034
RACE×SEX	14/18	0.678	0.614	0.059	0.611	0.517	0.099	0.429	0.410	0.155
RACE×DIS	13/18	0.697	0.634	0.000	0.686	0.554	0.000	0.544	0.464	0.000
RACE×SEX×DIS	21/35	0.746	0.683	0.000	0.711	0.580	0.000	0.600	0.519	0.000

Aggregate audits hide most of the disparity. For the BN, ΔEOpp is 0.162 by sex and 0.384 by disability, 0.468 for sex×disability, and **0.746 intersectionally**. Every disabled intersection has a TPR of at most 0.08. Disabled low-income adults mostly *do* have public coverage, so the model ranks them low, and the minority of them without coverage would almost never be reached. Survey weighting does not change this. The weighted test TPR is 0.011 (SE 0.006) for disabled men and 0.031 (0.009) for disabled women, against 0.280 (0.006) and 0.448 (0.006) for men and women without a disability (80 successive-difference replicate weights). Texas 2018 replicates the pattern: 0.094 by sex, 0.390 by disability, 0.432 for sex×disability, 0.888 intersectionally.

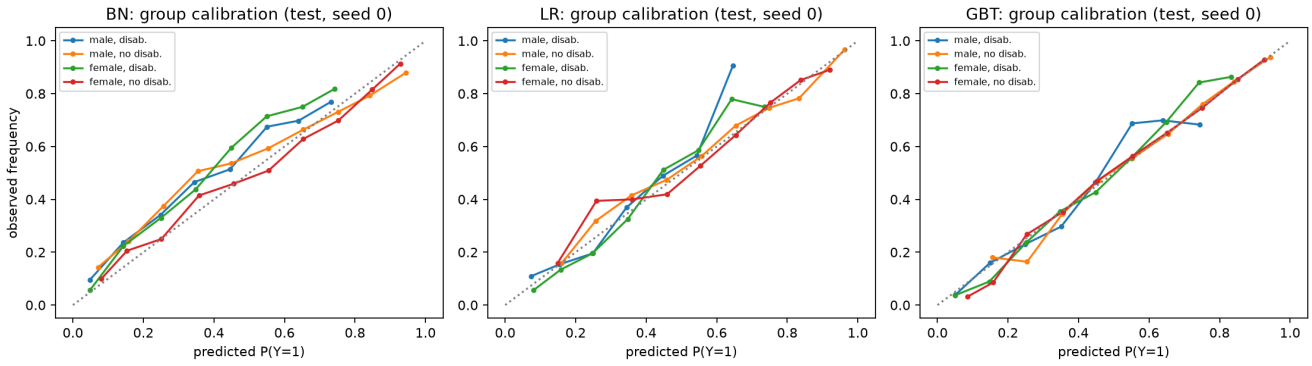


Figure 3: Group calibration by sex×disability (test, seed 0).

6.3 Audit-model behaviour on real counts

Table 8 (appendix) lists every intersection with its denominators. Pooling moves estimates appreciably only for groups with 1–7 positives (pull 0.05–0.14), and no group with ≥30 positives moves by more than 0.032 (Figure 4). On these counts the struct model fits $\hat{\tau} = 0.01$ (its lower bound) and reduces to a main-effects logistic model. It gives an implausibly narrow 90% interval for ΔEOpp, (0.750, 0.757), against (0.729, 0.816) for hier. Every audit model gives $P(\Delta\text{EOpp} > 0.10 \mid \text{data}) = 1.0$ for the unconstrained rule, at every tolerance and prior tested (notebook §12).

6.4 Fairness-adjusted rules selected on validation

Every validation-time "tolerance satisfied" claim was false on test. The pooled criteria (hier, full Bayes, complete pooling, the calibrated proposed rule, and struct in 2 of 5 splits) found a λ meeting $\epsilon = 0.10$ on validation. Across all splits and thinning levels, all 116 such claims had a test ΔEOpp between 0.227 and 0.770. Raw and independent criteria never made the claim. The utility cost of the most fairness-seeking rule ($\lambda = 1$) is 0.008 recall on average (at most 0.013 in any split), within the 0.02 budget. Figure 5 shows the trade-off curves. In the confirmatory tests, the calibrated rule and hier plug-in chose the same λ at seed 0 ($\Delta = 0$). Against raw plug-in the difference is -0.052, 95% CI $[-0.194, 0.051]$, Holm $p = 0.736$. Across the 5 refits the sign of the difference changes

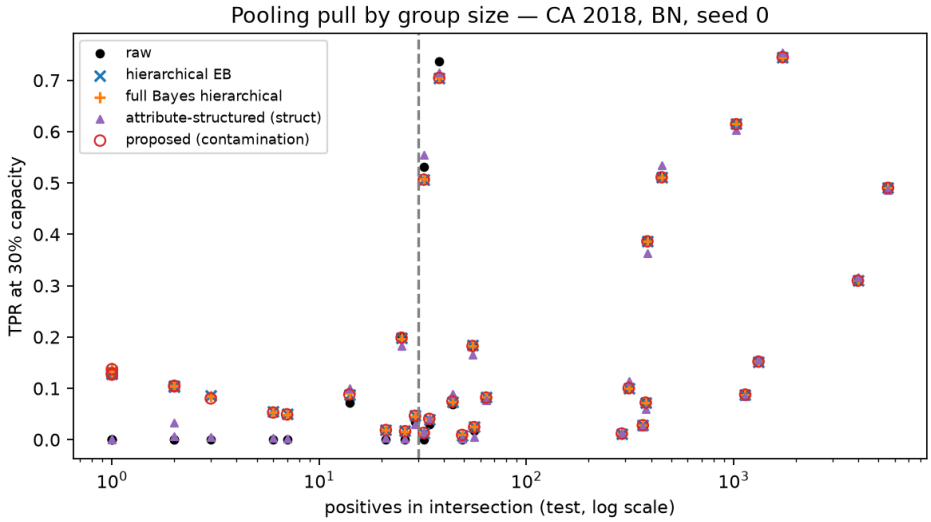


Figure 4: Raw and pooled TPR estimates by intersection size (BN, seed 0, test).

Table 3: Intact training data, 5 household splits. "Feasible" = the criterion found a λ with value $\leq \epsilon$ on validation. The test $P(\Delta > \epsilon)$ comes from the calibrated contamination audit on test counts.

Selection criterion	feasible on val.	mean λ	test recall	test sel. rate	test ΔEOpp (mean $\pm$ sd)	worst recall	test $P(\Delta > \epsilon)$
raw · plug-in	0/5	0.80	0.385	0.299	0.348 ± 0.077	0.172	0.997
independent · upper	0/5	0.84	0.383	0.299	0.418 ± 0.094	0.191	0.996
complete · plug-in	5/5	0.26	0.390	0.298	0.641 ± 0.066	0.019	1.000
hier EB · plug-in	5/5	0.90	0.383	0.299	0.323 ± 0.077	0.210	0.965
hier EB · upper	5/5	1.00	0.380	0.299	0.367 ± 0.060	0.260	0.952
full Bayes · upper	5/5	1.00	0.380	0.299	0.367 ± 0.060	0.260	0.950
struct · plug-in	2/5	1.00	0.380	0.299	0.367 ± 0.060	0.260	0.955
struct · upper	2/5	1.00	0.380	0.299	0.367 ± 0.060	0.260	0.950
proposed · plug-in (calibrated)	5/5	0.96	0.381	0.299	0.332 ± 0.072	0.245	0.921
proposed · upper	0/5	1.00	0.380	0.299	0.367 ± 0.060	0.260	0.952

(Table B). Out of sample on Texas 2018 (5 splits), pooled criteria made 27 validation-time tolerance claims and all 27 failed on the Texas test split (test ΔEOpp 0.110–0.888), supporting the falsifiable claim H1 (notebook §14.1).

6.5 Audit procedure: simulation evidence (SYNTHETIC and SEMI-SYNTHETIC)

The calibrated procedure has FTS 0.503 and ACC 0.830 on the synthetic calibration design. On the real design its FTS is 0.133–0.280, but **its ACC drops to 0.32**. The calibration is design-specific and must be redone on the deployment design. Exchangeable pooling (hier) declares the 24-positive exception "fair" in 98–100% of worlds, and the attribute-structured audit masks it whenever the deficit is an interaction. Every estimator detects a deficit carried by the disability main effect.

6.6 Improved procedure: design-matched calibration and held-out certification (notebook §15)

Preregistered hypotheses H2a–c were tested out of sample on the Texas design (Table 6, Figure 7). The design-matched audit scores $J = \text{ACC} - \text{mean FTS} = 0.664$, against at most 0.385 for every established audit and 0.189 for the invented-design calibration. Its false-certification rate is 0.168, against 0.615–0.661 for hierarchical and full-Bayes pooling (H2a, H2b supported). On the California design, which the pilot used, it only ties the pooled audits on J (0.330 vs 0.333–0.334), with half their false-certification rate. Held-out certification made no false tolerance claims on real CA or TX data, against 10 of 10 false claims with same-data certification, at unchanged recall (Table 7; H2c supported).

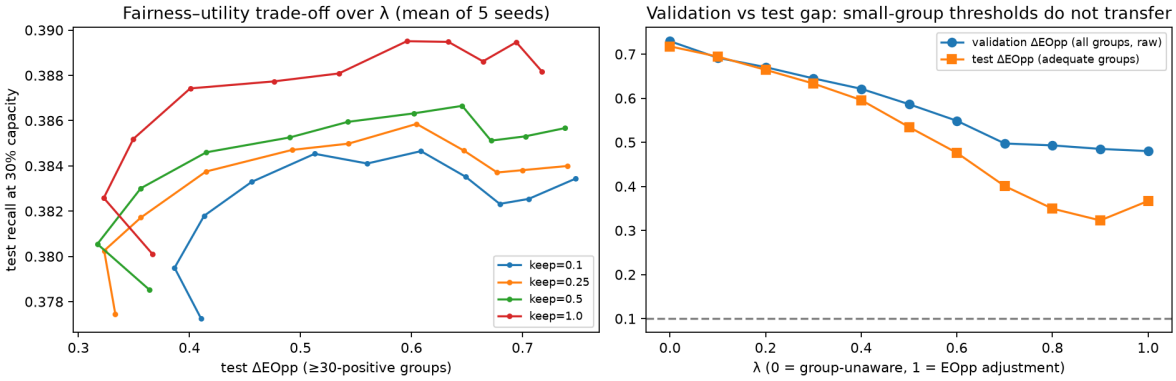


Figure 5: Fairness–utility trade-off over λ and validation vs test gaps (mean of 5 splits).

Table 4: SYNTHETIC (notebook §8): 16 groups of 4000 to 8 positives, one 20-positive group 0.20 below the rest, $R = 300$, $\epsilon = 0.10$.

	FTS (exceptional group kept)			ACC	90% gap interval	
Estimator · rule	100%	50%	25%	homog.	coverage	width
raw · plugin	0.003	0.000	0.000	0.007	–	–
independent · upper 0.9	0.000	0.000	0.000	0.000	0.264	0.382
complete · plugin	1.000	1.000	1.000	1.000	0.000	0.000
hier · plugin	1.000	1.000	1.000	1.000	0.000	0.018
hier · upper 0.9	0.980	0.987	0.987	0.993	0.000	0.018
fullbayes · plugin	0.983	1.000	0.993	1.000	0.038	0.114
fullbayes · upper 0.9	0.310	0.337	0.397	0.427	0.038	0.114
proposed $\pi \leq$0.8 · plugin (calibrated)	0.503	0.593	0.563	0.830	0.962	0.399
proposed $\pi \leq$0.8 · upper 0.5	0.450	0.567	0.577	0.773	0.962	0.399
proposed $\pi \leq$0.8 · upper 0.9	0.000	0.000	0.000	0.000	0.962	0.399
proposed $\pi \leq$0.95 · upper 0.9	0.297	0.300	0.317	0.527	0.589	0.273
proposed $\pi \leq$0.99 · upper 0.9	0.780	0.830	0.883	0.930	0.200	0.114

7 Ethical considerations

- **Systematic exclusion of disabled people:** the unconstrained rule almost never selects disabled people without public coverage. Any real outreach built this way would miss them.
- **Pooling can erase small groups:** a hierarchical audit can report a small disadvantaged intersection as average. Report raw counts and the pooling pull next to every pooled estimate.
- **Who chooses ϵ and the capacity:** the analysts chose $\epsilon = 0.10$ and 30% capacity; they were not set by affected communities. No stakeholder input was obtained: no affected community, outreach agency or disability organisation was consulted on ϵ, the capacity or the label. This is a limitation, and both values are reported with sensitivity analyses ($\epsilon \in \{0.05, 0.10, 0.15\}$).
- **Label meaning:** absence of public coverage is not uninsurance (people may hold private coverage) and is not eligibility. Mislabelling these concepts could misdirect resources.
- **Recorded categories:** race and disability categories are survey constructs. Keeping all nine race codes preserves their meaning but creates very small cells.

8 Caveats and recommendations

- No rule in the family reaches ΔEOpp ≤ 0.10 on test; the smallest observed value is 0.227. **Do not describe any configuration as fair.**

Table 5: SEMI-SYNTHETIC (notebook §9.2): the real seed-0 validation design (32 groups, 1 to 5,499 positives), invented TPRs. The deficit is at `2|2|1` (Black, female, disabled; 24 positives), thinned to 25% in the "keep .25" columns. $R = 300$.

	ACC	FTS cell		FTS interaction		FTS dis.	$\theta_{2\|2\|1}$ (cell, keep 1)	
Procedure	keep 1	keep 1	keep .25	keep 1	keep .25	keep 1	90% cov.	post. mean
proposed (robust@0.8, plugin)	0.323	0.133	0.280	0.160	0.233	0.000	0.720	0.281
fullbayes · plugin	0.997	1.000	0.997	1.000	1.000	0.000	0.000	0.371
fullbayes · upper0.9	0.413	0.397	0.477	0.520	0.540	0.000	0.000	0.371
hier · plugin	1.000	1.000	0.997	1.000	1.000	0.000	0.000	0.378
hier · upper0.9	0.970	0.997	0.983	0.993	0.987	0.000	0.000	0.378
independent · plugin	0.000	0.000	0.000	0.000	0.000	0.000	0.900	0.201
independent · upper0.9	0.000	0.000	0.000	0.000	0.000	0.000	0.900	0.201
raw · plugin	0.000	0.000	0.000	0.000	0.000	0.000	–	–
struct · plugin	0.277	0.293	0.367	0.367	0.363	0.000	0.000	0.368
struct · upper0.9	0.217	0.233	0.327	0.320	0.313	0.000	0.000	0.368

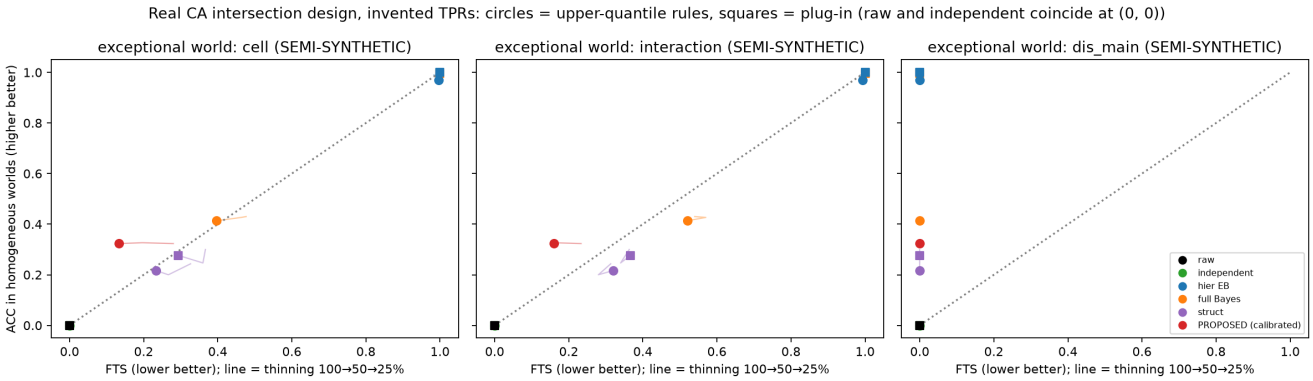


Figure 6: SEMI-SYNTHETIC tolerance-decision frontier on the real CA design.

- Do not certify a tolerance from a pooled estimate when any intersection has fewer than about 30 positives. Validation-time claims did not transfer to test.
- Bayesian credible intervals are not frequentist guarantees. The EB struct audit covers 0.665 at nominal 0.90 even under its own model (oracle 0.905), and exchangeable pooling gives near-zero coverage for the gap in simulation.
- Calibrate the audit procedure on the design it will be used on, and certify on data not used to choose the rule. The invented-design calibration lost acceptance (ACC 0.83 → 0.32), while design-matched calibration reached $J = 0.664$ out of sample; same-data certification produced 10 of 10 false claims, held-out certification none.
- Results are for the unweighted benchmark sample unless stated. The survey-weighted estimates cover the sex×disability groups only.
- **Relation to prior work.** Hierarchical Bayesian estimation of intersectional fairness is not ours (Foulds et al. 2020); our `struct` audit is a restricted version in that spirit, not an exact reproduction. That marginal fairness can hide subgroup unfairness is known (Kearns et al. 2018); our interaction-only world is an instance. That fairness constraints met on training data can fail out of sample is known (Cotter et al. 2019). Our contribution is the evaluation of pooled *audit criteria* under sparsity and the design-matched, held-out audit protocol, not a new prior or metric.

Table 6: Out-of-sample test on the Texas design (SEMI-SYNTHETIC; $R = 200$ per cell).

Procedure	ACC (fair)	FTS cell	FTS interaction	FTS dis. main	mean FTS	J	90% gap coverage
design-matched (centred-slab contam., $\pi \leq 0.95 \cdot$ upper 0.5)	0.832	0.255	0.250	0.000	0.168	**0.664**	0.615
fullbayes · plugin	1.000	0.920	0.925	0.000	0.615	**0.385**	0.174
hier · upper 0.9	0.985	0.930	0.945	0.000	0.625	**0.360**	0.013
hier · plugin	1.000	0.990	0.992	0.000	0.661	**0.339**	0.013
fullbayes · upper 0.9	0.290	0.110	0.145	0.000	0.085	**0.205**	0.174
§9 procedure (contam., $\pi \leq 0.8 \cdot$ plugin)	0.218	0.048	0.038	0.000	0.028	**0.189**	0.506
struct · plugin	0.108	0.125	0.095	0.000	0.073	**0.034**	0.037
struct · upper 0.9	0.090	0.105	0.078	0.000	0.061	**0.029**	0.037
independent · upper 0.9	0.000	0.000	0.000	0.000	0.000	**0.000**	0.000
raw · plugin	0.000	0.000	0.000	0.000	0.000	**0.000**	–

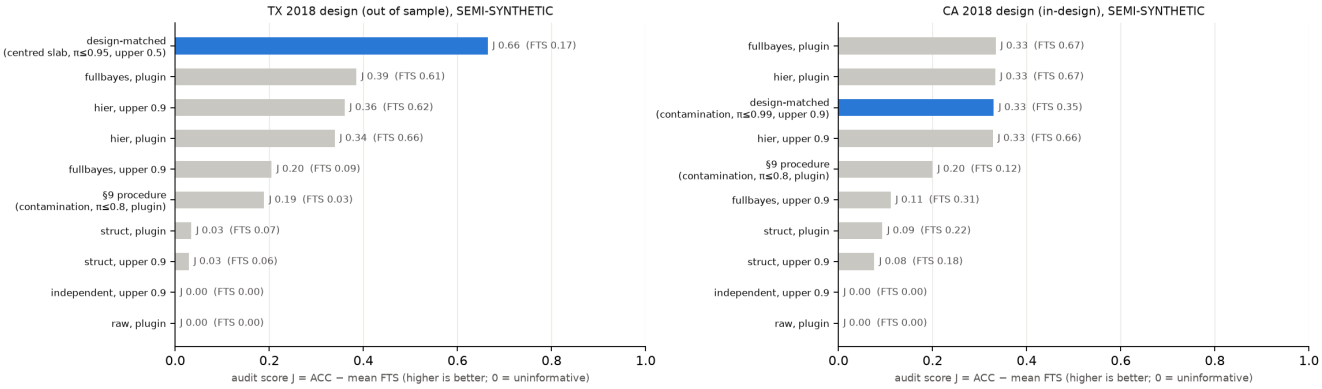


Figure 7: Audit score J per procedure on the Texas (out of sample) and California designs.

Table 7: Real data: tolerance claims of the design-matched procedure, same-data vs held-out certification.

State	certification	claims ($\leq \epsilon$)	false claims	mean test ΔEOpp	mean test recall
CA	same-data	5/5	5	0.367	0.380
CA	held-out	0/5	0	0.351	0.379
TX	same-data	5/5	5	0.238	0.350
TX	held-out	0/5	0	0.267	0.350

A Intersectional audit table

Table 8: Intersectional audit table for the BN rule (unconstrained top-30%, seed-0 test split). n = people, n^+ = positives (no public coverage), k = selected positives, TPR = k/n^+. Posterior means under each audit model; pull = hier − raw. The contamination column uses $\pi_{\max} = 0.95$ because this table is computed (notebook §7) before the §9 calibration fixes $\pi_{\max} = 0.8$. $*$ = exploratory (< 30 positives or < 30 negatives).

Code	Race (RAC1P)	Sex	Disab.	n	n^+	k	raw TPR	indep.	hier	struct	contam. $\pi \leq 0.95$	pull
7|2|1	NHPI	F	yes	4	0	0	–	–	–	–	–	–*
4|2|2	Alaska Native	F	no	1	1	0	0.000	0.324	0.129	0.000	0.137	0.129*
4|2|1	Alaska Native	F	yes	2	1	0	0.000	0.330	0.130	0.000	0.130	0.130*
4|1|1	Alaska Native	M	yes	1	1	0	0.000	0.337	0.132	0.000	0.127	0.132*
5|2|1	AIAN, tribe spec./n.s.	F	yes	9	2	0	0.000	0.248	0.105	0.006	0.104	0.105*
7|1|1	NHPI	M	yes	8	2	0	0.000	0.252	0.103	0.033	0.105	0.103*
3|2|1	American Indian	F	yes	16	3	0	0.000	0.197	0.085	0.005	0.080	0.085*
5|1|1	AIAN, tribe spec./n.s.	M	yes	14	6	0	0.000	0.124	0.055	0.003	0.053	0.055*
3|1|1	American Indian	M	yes	20	7	0	0.000	0.110	0.050	0.003	0.049	0.050*
5|1|2	AIAN, tribe spec./n.s.	M	no	24	14	1	0.071	0.127	0.088	0.100	0.088	0.017*
2|2|1	Black	F	yes	142	21	0	0.000	0.044	0.019	0.003	0.019	0.019*
5|2|2	AIAN, tribe spec./n.s.	F	no	37	25	5	0.200	0.221	0.199	0.183	0.199	−0.001*
2|1|1	Black	M	yes	187	26	0	0.000	0.036	0.017	0.002	0.017	0.017*
9|2|1	Two or more	F	yes	77	29	1	0.034	0.064	0.047	0.030	0.047	0.013*
9|1|1	Two or more	M	yes	97	32	0	0.000	0.030	0.013	0.015	0.013	0.013
7|1|2	NHPI	M	no	48	32	17	0.531	0.528	0.506	0.554	0.506	−0.025*
6|1|1	Asian	M	yes	125	34	1	0.029	0.056	0.040	0.040	0.041	0.011
7|2|2	NHPI	F	no	57	38	28	0.737	0.725	0.704	0.716	0.705	−0.032*
3|1|2	American Indian	M	no	84	44	3	0.068	0.088	0.074	0.089	0.075	0.006
8|1|1	Some other race	M	yes	209	49	0	0.000	0.019	0.009	0.002	0.009	0.009
3|2|2	American Indian	F	no	89	55	10	0.182	0.192	0.183	0.165	0.183	0.002
8|2|1	Some other race	F	yes	197	56	1	0.018	0.035	0.025	0.005	0.024	0.007
6|2|1	Asian	F	yes	159	64	5	0.078	0.091	0.083	0.077	0.082	0.005
1|1|1	White	M	yes	965	289	3	0.010	0.014	0.012	0.012	0.012	0.002
2|2|2	Black	F	no	541	312	31	0.099	0.102	0.100	0.114	0.100	0.001
1|2|1	White	F	yes	1046	365	10	0.027	0.030	0.028	0.025	0.028	0.001
2|1|2	Black	M	no	655	377	27	0.072	0.074	0.073	0.060	0.072	0.001
9|1|2	Two or more	M	no	547	384	149	0.388	0.389	0.387	0.362	0.386	−0.001
9|2|2	Two or more	F	no	643	450	231	0.513	0.513	0.512	0.534	0.511	−0.001
6|1|2	Asian	M	no	1488	1030	634	0.616	0.615	0.615	0.602	0.614	0.000
8|1|2	Some other race	M	no	1824	1139	100	0.088	0.088	0.088	0.084	0.088	0.000
8|2|2	Some other race	F	no	2273	1320	201	0.152	0.153	0.152	0.156	0.152	0.000

Code	Race (RAC1P)	Sex	Disab.	n	n^+	k	raw TPR	indep.	hier	struct	contam. $\pi \leq 0.95$	pull
6|2|2	Asian	F	no	2283	1727	1288	0.746	0.745	0.745	0.753	0.745	−0.001
1|1|2	White	M	no	5805	3980	1233	0.310	0.310	0.310	0.317	0.310	0.000
1|2|2	White	F	no	7976	5570	2734	0.491	0.491	0.491	0.486	0.491	0.000

B Confirmatory and exploratory paired bootstrap (seed 0, $B = 1000$)

Comparison	family	metric	estimate	95% CI	p_{boot}	p_{Holm}
calibrated − hier plug-in	confirmatory	ΔEOpp	+0.0000	[+0.0000, +0.0000]	1.000	1.000
calibrated − hier plug-in	confirmatory	recall	+0.0000	[+0.0000, +0.0000]	1.000	–
calibrated − raw plug-in	confirmatory	ΔEOpp	−0.0517	[−0.1944, +0.0505]	0.368	0.736
calibrated − raw plug-in	confirmatory	recall	−0.0061	[−0.0089, −0.0035]	0.001	–
proposed upper − hier plug-in	exploratory	ΔEOpp	−0.0714	[−0.1755, +0.0000]	0.074	–
proposed upper − hier plug-in	exploratory	recall	−0.0026	[−0.0048, −0.0005]	0.014	–
proposed upper − raw plug-in	exploratory	ΔEOpp	−0.1231	[−0.3073, +0.0091]	0.076	–
proposed upper − raw plug-in	exploratory	recall	−0.0086	[−0.0124, −0.0053]	0.001	–
struct upper − hier plug-in	exploratory	ΔEOpp	−0.0714	[−0.1755, +0.0000]	0.074	–
struct upper − hier plug-in	exploratory	recall	−0.0026	[−0.0048, −0.0005]	0.014	–
struct upper − raw plug-in	exploratory	ΔEOpp	−0.1231	[−0.3073, +0.0091]	0.076	–
struct upper − raw plug-in	exploratory	recall	−0.0086	[−0.0124, −0.0053]	0.001	–
struct plug-in − hier plug-in	exploratory	ΔEOpp	−0.0714	[−0.1755, +0.0000]	0.074	–
struct plug-in − hier plug-in	exploratory	recall	−0.0026	[−0.0048, −0.0005]	0.014	–
struct plug-in − raw plug-in	exploratory	ΔEOpp	−0.1231	[−0.3073, +0.0091]	0.076	–
struct plug-in − raw plug-in	exploratory	recall	−0.0086	[−0.0124, −0.0053]	0.001	–

The confirmatory family (Holm) is the calibrated procedure vs hier plug-in and vs raw plug-in, on test ΔEOpp. All other rows are exploratory. Training uncertainty (5 refits): calibrated − raw plug-in = -0.016 ± 0.039 (signs $- + -0-$); calibrated − hier plug-in = $+0.009 \pm 0.035$ (signs $0 + + - 0$).